

Mehul Damani

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Education

- Ph.D. Massachusetts Institute of Technology (MIT)** Aug 2022 – present
Ph.D. candidate in Computer Science
 • **Advisor:** [Jacob Andreas](#) ; **GPA:** 4.9/5.0
- B.Eng Nanyang Technological University (NTU)** Aug 2018 – May 2022
Bachelor's in Mechanical Engineering
 • **GPA:** 4.74/5.0 ; Honours (Highest Distinction)

Experience

- MIT-IBM Watson AI Lab** Cambridge, MA
Research Intern
 May 2025 – Aug 2025
 • Reinforcement Learning for LLM tool use.
- New York University** Remote
Research Intern (Advised by [Lerrel Pinto](#))
 Jan 2021 – Aug 2022
 • Developed curriculum generation method that improved performance by upto 20% on robotic tasks.
- National University of Singapore** Singapore
Research Intern (Advised by [Guillaume Sartoretti](#))
 Apr 2020 – July 2022
 • Developed decentralized reinforcement learning (RL) methods for multi-agent systems.
 • Co-authored 4 papers, open-sourced code with 100+ stars on Github.
 • Achieved fourth position in the RL track of the NeurIPS 2020 Flatland Challenge.
- Satellite Research Centre (NTU)** Singapore
Research Intern
 Sept 2019 – Apr 2020
 • Developed regression models to characterize drift and bias of sensors for integration into ADCS of a satellite.

Publications

- [12] **Vector Policy Optimization: Training for Diversity Improves Test-Time Search** 2026
 Ryan Bahlous-Boldi, Isha Puri, Idan Shenfeld, Akarsh Kumar, **Mehul Damani**, Sebastian Risi, Omar Khattab, Zhang-Wei Hong, Pulkit Agrawal
Neural Information Processing Systems (NeurIPS)
- [14] **Right in the Right Way: Combining Verifiable Rewards with Human Demonstrations** 2026
Mehul Damani, Isha Puri, Idan Shenfeld, Jacob Andreas
ICML 2026 Workshop on RL from World Feedback
- [13] **Self-Distillation Enables Continual Learning** 2026
 Idan Shenfeld, **Mehul Damani**, Jonas Hubotter, Pulkit Agrawal
International Conference on Machine Learning (ICML) (Spotlight)
ICLR 2026 Workshop on Lifelong Agents (LLA) (Oral)
- [15] **Position: It's Time to Optimize for Self-Consistency** 2026
 Itamar Pres, Belinda Z. Li, Laura Ruis, Zifan Carl Guo, Keya Hu, **Mehul Damani**, Isha Puri, Ekdeep Singh Lubana, Jacob Andreas
International Conference on Machine Learning (ICML)
- [11] **Escaping the Mode: Multi Answer Reinforcement Learning in LMs** 2026
 Isha Puri, **Mehul Damani**, Idan Shenfeld, Marzyeh Ghassemi, Jacob Andreas, Yoon Kim

- [10] **Beyond Binary Rewards: Training LMs to Reason about their Uncertainty** 2026

Mehul Damani, Isha Puri, Stewart Slocum, Idan Shenfeld, Leshem Choshen, Yoon Kim, Jacob Andreas

International Conference on Learning Representations (ICLR)

COLM 2025 Workshop on Test-Time Scaling and Reasoning Models (Oral)

- [9] **The Surprising Effectiveness of Test-Time Training for Abstract Reasoning** 2025

Ekin Akyurek, **Mehul Damani**, Linlu Qiu, Han Guo, Yoon Kim, Jacob Andreas

International Conference on Machine Learning (ICML)

- [8] **Learning How Hard to Think: Input-Adaptive Allocation of LM Computation** 2025

Mehul Damani, Idan Shenfeld, Andi Peng, Andreea Bobu, Jacob Andreas

International Conference on Learning Representations (ICLR)

- [7] **Open Problems and Fundamental Limitations of Reinforcement Learning from Human Feedback** 2023

Stephen Casper, Xander Davies, Claudia Shi, Thomas Krendl Gilbert, Jérémy Scheurer, Javier Rando, Rachel Freedman, Tomasz Korbak, David Lindner, Pedro Freire, Tony Wang, Samuel Marks, Charbel-Raphaël Segerie, Micah Carroll, Andi Peng, Phillip Christoffersen, **Mehul Damani**, Stewart Slocum, Usman Anwar, Anand Siththaranjan, Max Nadeau, Eric J Michaud, Jacob Pfau, Dmitrii Krashennnikov, Xin Chen, Lauro Langosco, Peter Hase, Erdem Biyik, Anca Dragan, David Krueger, Dorsa Sadigh, Dylan Hadfield-Menell

Transactions on Machine Learning Research (TMLR)

- [6] **Mitigating Generative Agent Social Dilemmas** 2023

Julian Yocum, Phillip Christoffersen, **Mehul Damani**, Justin Svegliato, Dylan Hadfield-Menell, Stuart Russell

NeurIPS 2023 Foundation Models for Decision Making Workshop

- [5] **Formal Contracts Mitigate Social Dilemmas in Multi-Agent RL** 2023

Andreas Haupt, Phillip Christoffersen, **Mehul Damani**, Dylan Hadfield-Menell

Autonomous Agents and Multi-Agent Systems

- [4] **SocialLight: Distributed Cooperation Learning towards Network-Wide Traffic Signal Control** 2023

Harsh Goel, Yifeng Zhang, **Mehul Damani**, Guillaume Sartoretti

International Conference on Autonomous Agents and Multiagent Systems

- [3] **Distributed Reinforcement Learning for Robot Teams: a Review** 2022

Yutong Wang, **Mehul Damani**, Pamela Wang, Yuhong Cao, Guillaume Sartoretti

Current Robotics Reports, Springer

- [2] **PRIMAL2: Pathfinding Via Reinforcement and Imitation Multi-Agent Learning - Lifelong** 2021

Mehul Damani, Zhiyao Luo, Emerson Wenzel, Guillaume Sartoretti

IEEE Robotics and Automation Letters

- [1] **Flatland Competition 2020: MAPF and MARL for Efficient Train Coordination on a Grid World** 2020

Florian Laurent, Manuel Schneider, Christian Scheller, Jeremy Watson, Jiaoyang Li, Zhe Chen, Yi Zheng, Shao-Hung Chan, Konstantin Makhnev, Oleg Svidchenko, Vladimir Egorov, Dmitry Ivanov, Aleksei Shpilman, Evgenija Spirovska, Oliver Tanevski, Aleksandar Nikov, Ramon Grunder, David Galevski, Jakov Mitrovski, Guillaume Sartoretti, Zhiyao Luo, **Mehul Damani**, Nilabha Bhattacharya, Shivam Agarwal, Adrian Egli, Erik Nygren, Sharada Mohanty

NeurIPS 2020 Competition and Demonstration Track

Projects

Adversarial Vulnerabilities of CLIP

[Repo](#) 

- Generated interpretable, universal perturbations that caused misclassification of images by OpenAI's CLIP model.

MA-Sort

[Repo](#)

- Developed multi-agent RL benchmark to test emergence of dominance hierarchies in a group of RL agents.

Vigilant Bot

[Project Page](#)

- Developed RNN-based embedded device for detecting hand gestures and sending automated distress signals.

Vertical Take-off & Landing Aircraft (VTOL)

[Blog](#)

- Designed, assembled and tested an electric Vertical Take-off and Landing aircraft (VTOL) prototype.

Optimal Debris Deorbiting System

[Report](#)

- Conceptualized a method to deorbit space debris from low-earth orbit (LEO) using bidirectional ion thrusters.

Skills

Languages: Python, C, MATLAB

Maching Learning: TensorFlow, Torch, HuggingFace, Ray, WandB

Other: Conda, Docker, Git, Linux, Slurm

Honors and Awards

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| • Vicom Book Prize, <i>Top Scorer in Thermodynamics out of 500 students</i> | 2019 |
| • Dean's List, <i>Top 5% of cohort</i> | 2019, 2020 |
| • Kishore Vaigyanik Protsahan Yojana Fellowship, <i>Awarded by Department of Science and Technology, India</i> | 2017 |
| • National Talent Search Scholar, <i>Awarded by National Council on Educational Research, India</i> | 2017 |
| • Invited Tedx speaker on Black Holes and Time Travel | 2017 |

Reviewing

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| • Transactions on Machine Learning Research (TMLR) | 2025 |
| • Conference on Neural Information Processing Systems (NeurIPS) | 2022, 2024 |
| • International Conference on Learning Representations (ICLR) | 2023, 2025, 2026 |
| • International Conference on Robotics and Automation (ICRA) | 2021-2023 |
| • International Conference on Intelligent Robots and Systems (IROS) | 2023 |
| • International Conference on Machine Learning (ICML) | 2023, 2025, 2026 |
| • Association for the Advancement of Artificial Intelligence (AAAI) | 2022 |
| • Robotics and Automation Letters (RA-L) | 2022-2024 |

Invited Talks

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| • NVIDIA | October 2025 |
| • UT Austin | October 2025 |
| • CoLM SCALR Workshop Oral | October 2025 |
| • MIT AI Conference | October 2024 |

Teaching

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| • Teaching Assistant for 6.8300 (Advances in Computer Vision) | 2024 |
| • Peer Tutor for MA2007 (Thermodynamics) | 2021 |
| • Organized tutorial on Arduino-powered robotic cars for high-school students. | 2020 |